CHAPTER 7

Karnaugh Map (K-Map)

Draw · group · read the answer · SOP & POS · prime implicants · don't-cares

QUICK TRICK

K-map is the most important chapter for GATE. It is a picture puzzle: put 1s in boxes, circle neighbouring boxes, read the answer. The green boxes give you fast tricks.

7.1 What is a K-map and why do we need it?

● **LEVEL 1 · BEGINNER**

The problem. Simplifying with Boolean laws (Chapter 4) needs clever guessing. It is easy to miss a step.

The idea. Two terms that differ in only **one variable** can be joined, and that variable disappears:

```
A'BC + ABC = BC (A' + A) = BC          (A changed, so A disappears)
```

A K-map is a table where **neighbouring boxes differ in exactly one variable**. So joining neighbouring 1s does this simplification for us, by just looking.

Analogy. Think of children sitting in a grid. Neighbours who wear the same colour shirt except for one thing can form a team. Bigger teams mean a simpler answer.

7.2 The map: layout and Gray code order

● **LEVEL 1 · BEGINNER**

One box per minterm. A function of n variables has 2^n boxes. The small number in each box is its minterm number. The labels use **Gray code order: 00, 01, 11, 10** (never 00, 01, 10, 11), so that neighbours differ in **one bit**.

2 variables

A \ B	0	1
0	m0	m1
1	m2	m3

3 variables

A \ BC	00	01	11	10
0	m0	m1	m3	m2
1	m4	m5	m7	m6

4 variables

AB \ CD	00	01	11	10
00	m0	m1	m3	m2
01	m4	m5	m7	m6
11	m12	m13	m15	m14
10	m8	m9	m11	m10

QUICK TRICK

Gray code trick: “0, 1, 1, 0”. The order 00, 01, 11, 10 is the same as: low bit goes 0, 1, 1, 0 and the high bit goes 0, 0, 1, 1. Notice that the numbers jump: after m1 comes **m3**, and after m3 comes **m2**. Always read the number inside the box, not the position.

QUICK TRICK

The map is a donut. The left edge touches the right edge, and the top edge touches the bottom edge. So boxes on opposite edges are neighbours too. All four corners are neighbours of each other.

COMMON MISTAKE

Writing the labels as 00, 01, 10, 11. Then boxes that look like neighbours are not neighbours.

How to put a function on the map: write 1 in every box whose minterm number is in the Σ list, X in the don't-care boxes, and 0 everywhere else.

7.3 The grouping rules

● **LEVEL 2 · EXAM**

```
1) A group contains ONLY 1s (and X's if they help).
2) Group size must be a POWER OF 2: 1, 2, 4, 8, 16.
3) Groups are RECTANGLES (or squares). No diagonals, no L-shapes.
4) Groups can OVERLAP.
5) The map WRAPS around the edges (donut).
6) Make groups as BIG as possible.
7) Use as FEW groups as possible.
```

Group size	Name	Variables that disappear	Literals left (n = 4)
1	Single	0	4
2	Pair	1	3
4	Quad	2	2
8	Octet	3	1
16	All	4	0 (the answer is 1)

QUICK TRICK

Size trick: a group of 2^k boxes removes **k** variables. Literals left = n − k. Example: in 4 variables a quad (k = 2) gives a term with 4 − 2 = 2 literals.

COMMON MISTAKE

Groups of 3, 5, 6 or 7 are **not allowed**. Split them: 6 = 4 + 2 (overlapping is fine).

Words you must know

Word	Meaning
Implicant	Any valid group of 1s.
Prime implicant (PI)	A group that cannot be made any bigger.
Essential prime implicant (EPI)	A PI that covers a 1 that no other PI covers. **You must use every EPI.**
Redundant group	A group whose 1s are all covered by other groups you already chose. **Do not use it.**

7.4 How to read the answer from a group

● LEVEL 2 · EXAM

For each group, look at every variable. If it **never changes** inside the group, keep it. If it **changes** (is both 0 and 1), drop it.

In the group the variable is…	SOP (group the 1s)	POS (group the 0s)
always 1	write the letter plain (A)	write the letter with a bar (A')
always 0	write the letter with a bar (A')	write the letter plain (A)
changes	leave it out	leave it out
Join the pieces	AND the letters inside a group, OR the groups	OR the letters inside a group, AND the groups

QUICK TRICK
SOP: 1 → plain, 0 → bar, AND them, OR the groups.
POS: 0 → plain, 1 → bar, OR them, AND the groups.

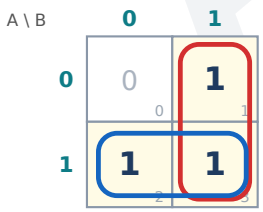
The steps (follow this every time)

```
STEP 1  Draw the map. Write 1s (and X's, 0s).
STEP 2  Find LONELY 1s (no neighbour). They stay as single boxes.
STEP 3  Find 1s that can be grouped in ONLY ONE way. That group is essential.
STEP 4  Take the BIGGEST groups first (octet, quad, pair). Use edges and corners.
STEP 5  Make sure EVERY 1 is covered. Use the fewest extra groups.
STEP 6  Read each group (keep letters that do not change).
STEP 7  Write the answer: OR the terms (SOP)   or   AND the terms (POS).
```

7.5 Worked examples

Example 1 — two variables (warm-up) · Level 1

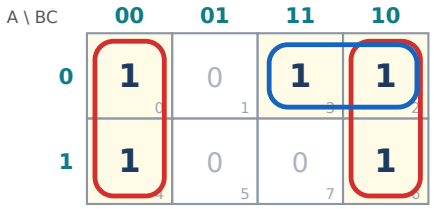
F = Σm(1, 2, 3)



Group 1 (red): m1, m3 → column B = 1 → term **B**.
Group 2 (blue): m2, m3 → row A = 1 → term **A**.
Answer: F = A + B. (m3 is used twice. That is allowed.)

Example 2 — three variables with wrap-around · Level 2

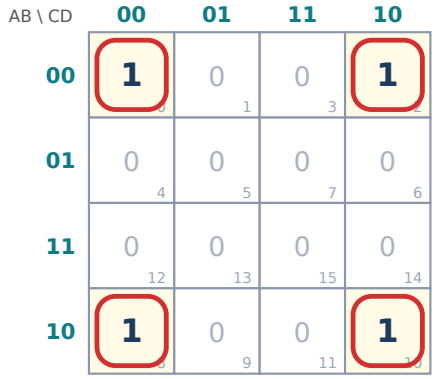
F = Σm(0, 2, 3, 4, 6)



Red quad m0, m2, m4, m6: the columns BC = 00 and 10 are on opposite edges, but they are neighbours (donut). In this group C is always 0 → **C'**. A and B change → dropped.
Blue pair m2, m3: A = 0 and B = 1 stay the same, C changes → **A'B**.
Answer: F = C' + A'B.

Example 3 — the four corners · Level 2

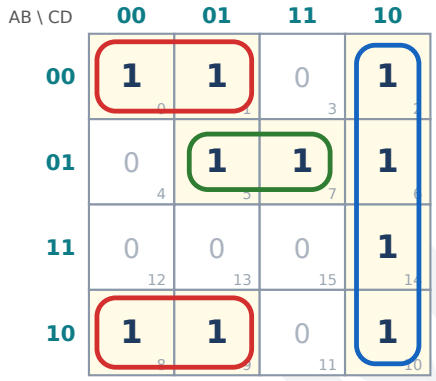
F = Σm(0, 2, 8, 10)



The four corners are neighbours (left↔right and top↔bottom), so they form **one quad**.
In the group B = 0 always and D = 0 always. A and C change.
Answer: F = B'D'.

Example 4 — four variables, redundant group · Level 3

F = Σm(0, 1, 2, 5, 6, 7, 8, 9, 10, 14)

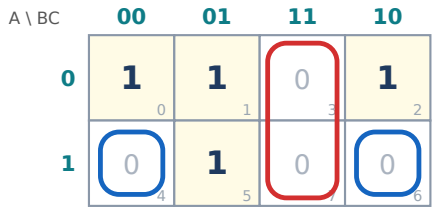


Blue quad m2, m6, m10, m14 = whole column CD = 10 → C = 1, D = 0 → **CD'**. m14 is covered only here, so it is **essential**.
Red quad m0, m1, m8, m9 (top and bottom rows joined) → B = 0, C = 0 → **B'C'**. m9 is covered only here, so it is **essential**.
Green pair m5, m7 covers the two remaining 1s → A = 0, B = 1, D = 1 → **A'BD**.
Answer: F = CD' + B'C' + A'BD.
The corner quad B'D' (m0, m2, m8, m10) is a prime implicant, but its 1s are already covered by the red and blue groups, so it is **redundant**. Do not add it.

Prime implicant	Cells	Type
CD'	2, 6, 10, 14	Essential (only one covering m14)
B'C'	0, 1, 8, 9	Essential (only one covering m9)
A'BD	5, 7	Needed: it covers m5 and m7 with ONE term
B'D'	0, 2, 8, 10	Redundant (not needed)
A'C'D, A'BC	1+5, 6+7	Not chosen: you would need two terms to cover m5 and m7

Example 5 — POS: group the 0s · Level 2

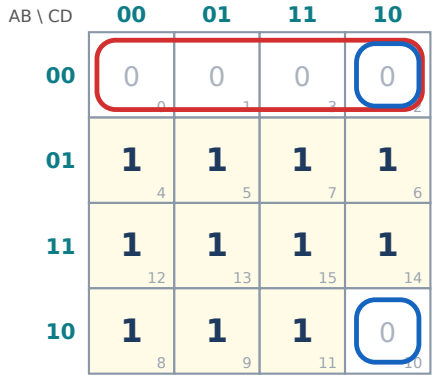
F = Σm(0, 1, 2, 5) (find POS)



Zeros are m3, m4, m6, m7. Group the **0s**.
Red pair m3, m7: B = 1, C = 1 always. POS rule: 1 → bar → sum term **(B' + C')**.
Blue pair m4, m6 (wraps across the edge): A = 1, C = 0 always. POS rule: 1 → bar, 0 → plain → **(A' + C)**.
Answer: F = (A' + C)(B' + C').

Example 6 — POS from a maxterm list · Level 3

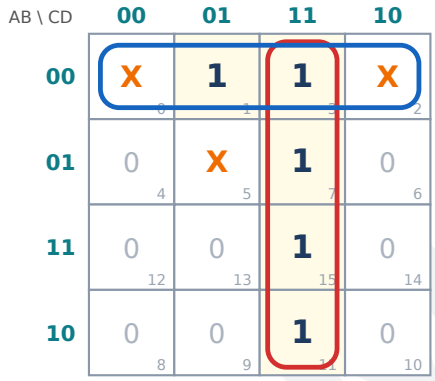
F = ΠM(0, 1, 2, 3, 10)



A maxterm list tells you the **0s**: m0, m1, m2, m3 and m10. All other boxes are 1.
Red quad (top row): A = 0, B = 0 → sum term **(A + B)**.
Blue pair m2, m10 (top and bottom joined): B = 0, C = 1, D = 0 → **(B + C' + D)**.
Answer: F = (A + B)(B + C' + D).

Example 7 — don't-cares · Level 3

F = Σm(1, 3, 7, 11, 15) + d(0, 2, 5)

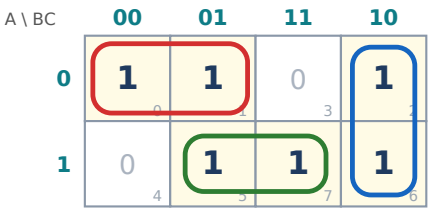


X means “we do not care”. Use an X as 1 **only if it makes a group bigger**. Never circle a group made only of X's.
Red quad m3, m7, m11, m15 (column CD = 11) → C = 1, D = 1 → **CD**.
Blue quad m0, m1, m2, m3 (top row, uses X at m0 and m2) → A = 0, B = 0 → **A'B'**.
Answer: F = A'B' + CD.
Another correct answer is **A'D + CD** (the quad m1, m3, m5, m7 uses the X at m5). Both have the same size.

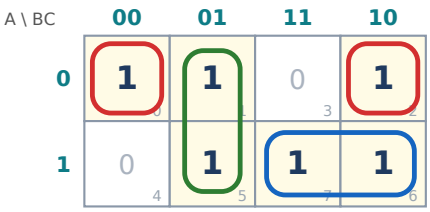
Example 8 — more than one correct answer · Level 3

F = Σm(0, 1, 2, 5, 6, 7). This map has six pairs and **no essential group**. Two different 3-group answers are correct:

Answer 1



Answer 2



Answer 1: F = A'B' + BC' + AC. **Answer 2: F = A'C' + AB + B'C.** Both are minimal. In GATE, check which one appears in the options.

7.6 Common mistakes

COMMON MISTAKE

- Wrong label order (00, 01, 10, 11).
- Groups of 3, 5, 6 (not powers of 2) or diagonal groups.
- Forgetting wrap-around and the four corners.
- Not making groups as large as possible.
- Adding a redundant group.
- Using the SOP reading rule for a POS group.
- Circling a group made only of don't-cares.

Quick-Solve Tricks — Summary

Situation	Trick
Labels	Always 00, 01, 11, 10 (low bit goes 0, 1, 1, 0).
Edges	Donut: left ↔ right, top ↔ bottom, 4 corners together.
Which group first?	Lonely 1s, then 1s with one choice, then biggest groups.
Group of 2^k boxes	Removes k variables.
Read a group (SOP)	Letter that never changes: 1 → plain, 0 → bar. Changing letters vanish.
Read a group (POS)	Group the 0s. Letter: 0 → plain, 1 → bar. Join with +, groups with ·
Maxterm list given	The listed numbers are the 0s. Group those.
Don't care	Use as 1 only to enlarge a group.
Whole map is 1 / 0	F = 1 / F = 0.
Check	Every 1 must be inside some group, and no group may contain a 0.
F' trick	Group the 0s as SOP to get F'. Then apply De Morgan to get the POS of F.

GATE QUICK CHECK — Chapter 7

```
n variables -> 2^n boxes          Order 00 01 11 10 (Gray)
Group size 2^k -> removes k variables
SOP : group 1s, keep constant letters (1 plain, 0 bar), AND inside, OR between
POS : group 0s, keep constant letters (0 plain, 1 bar), OR inside, AND between
EPI : covers a 1 that no other group covers (must be used)
```

Traps: label order · wrap-around · redundant groups · a minimal answer may not be unique · POS reading rule · don't-cares only to enlarge groups.

Practice questions

● **LEVEL 1 · BEGINNER**

Q1. How many boxes are in a 4-variable K-map?

Q2. Write the Gray order for two bits.

Q3. How many variables does a group of 8 boxes remove?

● **LEVEL 2 · EXAM**

Q4. Simplify F(A,B,C) = Σm(1, 3, 5, 7).

Q5. Simplify F(A,B,C) = Σm(0, 1, 4, 5).

Q6. Simplify F(A,B,C,D) = Σm(0, 2, 8, 10).

● LEVEL 3 · GATE

Q7. Find the minimal SOP of F(A,B,C) = Σm(0, 1, 2, 5, 6, 7).

Q8. Find the minimal SOP of F(A,B,C,D) = Σm(1, 3, 7, 11, 15) + d(0, 2, 5).

Q9. Find the minimal POS of F(A,B,C,D) = ΠM(0, 1, 2, 3, 10).

Q10. A 4-variable function is 1 only at minterms 5, 7, 13, 15. Simplify.

Q11. How many prime implicants does F = Σm(0, 1, 2, 5, 6, 7) (3 variables) have? How many are essential?

Q12. A 4-variable function is 1 for every minterm except m15. Find the minimal SOP.

Answers

Q	Solution
1	2^4 = **16** boxes.
2	**00, 01, 11, 10**.
3	2^k = 8 → k = **3** variables.
4	m1, m3, m5, m7 form a quad (columns BC = 01 and 11, both rows). Only C stays 1 → **F = C**.
5	m0, m1, m4, m5 form a quad (columns BC = 00 and 01, both rows). B = 0 always → **F = B'**.
6	Four corners form one quad. B = 0, D = 0 → **F = B'D'**.
7	No essential groups. Take three pairs: **F = A'B' + BC' + AC** (or A'C' + AB + B'C).
8	Quad m3, m7, m11, m15 = CD. Quad m0, m1, m2, m3 (X at 0 and 2) = A'B'. **F = A'B' + CD** (or A'D + CD).
9	Zeros: m0–m3 (row AB = 00) → (A + B). Zeros m2, m10 → B = 0, C = 1, D = 0 → (B + C' + D). **F = (A + B)(B + C' + D)**.
10	m5 = 0101, m7 = 0111, m13 = 1101, m15 = 1111 form a quad. B = 1 and D = 1 stay the same → **F = BD**.
11	Six prime implicants (pairs): A'B', A'C', BC', AB, AC, B'C. **None** are essential (every 1 is covered by two pairs).
12	Four octets: A' (m0–m7), B' (every box with B = 0), C' and D'. m15 is the only box in none of them. **F = A' + B' + C' + D'** (this equals (ABCD)', a NAND).

Next: Chapter 8 — Quine–McCluskey (tabulation) method, with a step-by-step table and simple tricks.